

[Dashboard](#)[my courses](#)[AMCN SS 2022](#)[wireless transmission](#)[Quiz: Wireless Transmission - Signal Propagation](#)

Started at	Sunday, April 10, 2022 at 4:02 p.m
status	Completed
Finished at	Sunday, April 10, 2022 at 11:57 p.m
Spent time	7 hours 55 minutes
valuation	9.00 out of 9.00 (100 %)
feedback	Congratulations, excellent performance (90-100%).

question **1**

Complete

Score 1.00 out
of 1.00

In which scenarios would fast fading be experienced?

Choose one or more answers:

- ☐ a. Rain and snow
- ☒ b. Mobility of receiver
- ☒ c. Mobility of objects in the transmission path
- ☐ i.e. Modification of transmit power

question **2**

Complete

Score 1.00 out
of 1.00

What are the reasons for fast fading?

Choose one or more answers:

- ☒ a. Multiple transmission paths between sender and receiver
- ☐ b. Adaptation in modulation scheme
- ☒ c. Movement of sender
- ☐ i.e. Drift of carrier frequency

question **3**

Complete

Score 1.00 out
of 1.00

What exactly happens during fast fading?

Choose one or more answers:

- ☐ a. Signals from a sender travel over multiple paths to different senders resulting in an overlap at other senders
- ☐ b. Multiple transmission paths from different senders result in overlap of the signal at the receiver
- ☐ c. Multiple transmission paths result in the reception of the signal at multiple receivers
- ☒ i.e. Multiple transmission paths from a single sender result in overlap of the signal at the receiver



Frage 4

Vollständig

Erreichte
Punkte 1,00 von
1,00

What is the effect of small phase shifts between multiple versions of the arriving signal? Phase shifts of 360° between two equal strength multipath signals result in:

Wählen Sie eine oder mehrere Antworten:

- ☒ a. an increase in the signal strength
- ☐ b. no change
- ☐ c. a cancelation of the signal as a whole

Frage 5

Vollständig

Erreichte
Punkte 1,00 von
1,00

What is the effect of small phase shifts between multiple versions of the arriving signal? Phase shifts of 180° between two multipath signals with equal strength result in:

Wählen Sie eine oder mehrere Antworten:

- ☐ a. an increase in the signal strength
- ☐ b. no change
- ☒ c. a cancelation of the signal as a whole

Frage 6

Vollständig

Erreichte
Punkte 1,00 von
1,00

What is the effect of small phase shifts between multiple versions of the arriving signal? Phase shifts of 90° between two equal strength multipath signals results in:

Wählen Sie eine oder mehrere Antworten:

- ☒ a. an increase in the signal strength
- ☐ b. a decrease of the signal strength as a whole
- ☐ c. no change



Frage 7

Vollständig

Erreichte
Punkte 1,00 von
1,00

Why does fast fading depend on mobility?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Because mobility is a requirement for the presence of multiple paths
- ☒ b. Because fast fading is due to a shift in the phases between multiple paths which requires some dynamic behavior

Frage 8

Vollständig

Erreichte
Punkte 1,00 von
1,00

Why is fast fading frequency-dependent?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Because the exact phase depends on the frequency
- ☐ b. Because the number of paths depends on the frequency

Frage 9

Vollständig

Erreichte
Punkte 1,00 von
1,00

Why is it considered as the best case that the pathloss for line-of-sight connections is inversely proportional to the square of the distance between a transmitter and receiver?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. If the consumer remains the same, the energy behaves proportional to the square of the voltage. E.g. if voltage doubles, energy quadruples. Just pathloss is the square of the distance.
- ☒ b. Assuming an isotropic radiator and zero absorption in space, the sum of the energy at any radius of the bowl can be assumed the same independent of the size of the surface of the bowl itself. However, as the surface of the bowl is proportional to the square of its radius, the energy per any fixed area reduces proportional to the inverse squared distance.

Vorhergehende Aktivität

◀ Quiz: Drahtlose
Übertragung - Frequenz
und Signale

Direkt zu:



next activity

Quiz: Wireless Transmission
- Antennas ►



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